

Housing, Fertility, and Demographic Transition

Research update

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Abstract

How does a fertility decline change future housing demand, and can cheaper housing stabilize population? We add one feedback to the paper's existing lifecycle model: birth cohorts generate entrant households after twenty years, and those households add to housing demand. A two-generation model shows that a fall in the value of children produces a lower positive steady state only if the fertility response to cheaper housing is strong enough to restore replacement.

We calibrate the lifecycle model at a point along a dated transition rather than treating the 2023 cross-section as stationary. We construct a replacement benchmark, reweight its age distribution to 2007, impose the observed path of household totals and ages through 2023, and estimate ten parameters using twelve fertility, housing, tenure, and wealth moments evaluated along the simulated transition, principally at 2023. The model fits fertility stocks, tenure, and wealth more closely than housing quantities. Untargeted historical comparisons make clear which paths are explained and which are not. At the estimated terminal preferences, locally generated entry reaches at most 0.678 of replacement on the audited price grid, so no usable positive closed terminal root is verified over that range. We therefore distinguish a closed finite-horizon benchmark from an open population sensitivity. The exercise is a sequence of temporary equilibria, not a forecast or a perfect-foresight asset-price calculation.

1 The mechanism

Children raise the demand for space at the same ages when households have low liquid wealth and limited access to ownership. Large rental units are scarce, and buying a family-sized home requires a down payment. Older households, by contrast, own much of the large-housing stock and may retain it after their children leave because moving is costly and housing is part of the estate they plan to leave. The allocation of housing across generations can therefore raise the effective cost of a child even when the aggregate stock is large.

The feedback has two directions. First, a higher housing cost or a tighter access constraint reduces fertility. Second, children eventually form the next generation of households, so fertility changes future housing demand. The first direction is a household mechanism; the second is a demographic equilibrium condition. Combining them changes both the transition and the long-run response to fertility and housing shocks.

The existing model's intergenerational channel works as follows. Children change their parents' housing needs, while young and old households compete for the housing stock and face different tenure constraints. The bequest motive also affects old-age saving and housing retention. Estates are not, however, assigned to particular descendants: new entrants draw from a fixed wealth distribution. The present extension adds cohort-size accounting, not an inheritance-transfer mechanism.

These links have close precedents. Demographic effects on housing demand go back at least to Mankiw and Weil (1989), and recent evidence relates birth-cohort size to housing prices through entry into and exit from homeownership (Francke and Korevaar, 2025). Housing prices and mortgage access also affect childbearing (Lovenheim and Mumford, 2013; Dettling and Kearney, 2014; Hacamo, 2021). Malmberg

(2012) studies the two-way relation between cohort housing pressure and fertility, while Lee and Suh (2025) analyze a low-fertility housing transition. Our question is narrower: how does the housing–fertility transition change when tenure constraints and old-age retention affect childbearing, and births in turn determine the number of future households?

The next section presents the feedback with young and old households and no additional state variables. Section 3 then adds the same cohort accounting to the existing lifecycle model.

2 Two-generation theory

2.1 Environment

Young households. At date t , there are Y_t young households. Each has income y and chooses consumption c , housing for the adults h , and the number of children n . Preferences are

$$u(c, h, n) = c + \alpha \log h + v \log n, \quad \alpha > 0, \quad v > 0. \quad (1)$$

The parameter α governs the demand for adult housing and v governs the demand for children. Quasilinearity makes both demands available in closed form. We assume $y > \alpha + v$, so consumption is positive at the interior solution.

Child costs and young access. Each child requires $q \geq 0$ units of other goods and $a > 0$ units of housing. One unit of housing costs p . Obtaining family space also entails an access cost $\omega \geq 0$, which represents a small rental market, a down-payment constraint, or another tenure friction. The cost of a child is therefore $q + a(p + \omega)$.

Old households. There are O_t old households. They no longer choose fertility, and their housing demand is

$$h_O(p) = \frac{\bar{h}_O}{p}, \quad \bar{h}_O > 0. \quad (2)$$

The parameter $\bar{h}_O$ summarizes old households' desire to retain housing. A policy that encourages downsizing reduces $\bar{h}_O$. In the full model, housing retention follows from moving costs and warm-glow utility from estate wealth rather than from this demand schedule. The estate is not transferred to a particular model descendant.

Housing. Aggregate housing services are supplied according to

$$H^s(p) = \bar{H} p^\varepsilon, \quad \bar{H} > 0, \quad \varepsilon \geq 0. \quad (3)$$

The fixed-stock case is $\varepsilon = 0$.

Demography. The number $\bar{n} > 0$ is replacement fertility. After survival and household formation, $\bar{n}$ children produce one new young household.

In this static section, p is the current cost of one unit of housing services. The lifecycle model distinguishes that current rent, denoted r_t , from the asset price P_t that capitalizes future rents. They move proportionally only under a constant user cost and a stationary price path.

2.2 The young household

Taking p as given, a young household solves

$$\max_{c, h, n} c + \alpha \log h + v \log n \quad \text{subject to} \quad c + ph + [q + a(p + \omega)]n = y. \quad (4)$$

At an interior solution, the household chooses

$$n(p) = \frac{v}{q + a(p + \omega)}, \quad h_Y(p) = \frac{\alpha}{p} + an(p). \quad (5)$$

Thus a higher housing cost or a tighter access constraint reduces fertility. Young housing demand is adult housing, α/p , plus the space required by children, $an(p)$.

This is a stripped-down version of the household mechanism already in the paper. In the full model, the per-child room floor plays the role of a , the equivalence scale generates the nonhousing resource cost summarized by q , and the fertility-preference shifter plays the role of v . Tenure constraints and the small market for large rental units generate the friction summarized by ω . The parameter χ is instead the owner-service premium. The simple model removes heterogeneity, saving, and the rent-versus-own decision only to make the demographic feedback transparent.

2.3 Equilibrium and the steady state

Housing clearing adds young and old demand:

$$\bar{H}p_t^\varepsilon = Y_t h_Y(p_t) + O_t h_O(p_t). \quad (6)$$

For inherited cohort masses, the left side rises and the right side falls with p_t , so the clearing price is unique.

Children become the next young cohort, while the current young become old:

$$Y_{t+1} = \frac{n(p_t)}{\bar{n}} Y_t, \quad O_{t+1} = Y_t. \quad (7)$$

The ratio $n(p_t)/\bar{n}$ is the number of next-period young households produced by each current young household.

A steady state has a constant housing cost and constant positive cohort masses, so $n(p^*) = \bar{n}$ and $Y^* = O^*$.

Let $S_t = Y_t + O_t$ denote total adult-household population. In a steady state, $S^* = 2Y^* = 2O^*$.

Proposition 1 (Steady state). *A unique positive steady state exists if and only if $v > \bar{n}(q + a\omega)$. It is*

$$p^* = \frac{v/\bar{n} - q}{a} - \omega, \quad Y^* = O^* = \frac{\bar{H}(p^*)^{\varepsilon+1}}{\alpha + \bar{h}_O + a\bar{n}p^*}. \quad (8)$$

A lower value of children v or a higher access cost ω lowers both the steady-state housing cost and population. A higher $\bar{h}_O$ reduces population but does not change the steady-state housing cost. A proportional increase in housing supply $\bar{H}$ raises population in the same proportion and also leaves the steady-state housing cost unchanged.

The cohort law first determines the housing cost consistent with replacement, $n(p^*) = \bar{n}$. Housing clearing then determines how many households can live in the market at that cost. This is why a permanent increase in housing supply raises population rather than lowering the steady-state housing cost.

Even if market housing were free, a child would still cost $q + a\omega$. Hence cheaper housing cannot restore replacement when $v \leq \bar{n}(q + a\omega)$.

2.4 Comparative statics

Suppose the economy begins in a steady state, a permanent parameter change arrives after Y_t and O_t have been inherited, and the subsequent dynamics converge to the new steady state. The impact and steady-state effects are:

	Impact p_t	Impact n_t	Long-run p^*	Long-run n^*	Long-run population
Lower access cost, $\omega \downarrow$	+	+	+	0	+
Lower old housing demand, $\bar{h}_O \downarrow$	−	+	0	0	+
Greater housing supply, $\bar{H} \uparrow$	−	+	0	0	+

Lowering ω makes family housing easier to obtain. Fertility and housing demand rise immediately, so part of the change appears as a higher market price. Lower old housing demand and greater supply instead reduce the price on impact. In the new steady state, fertility is again $\bar{n}$. For the latter two reforms, the housing cost returns to its original level and the additional housing supports a larger population.

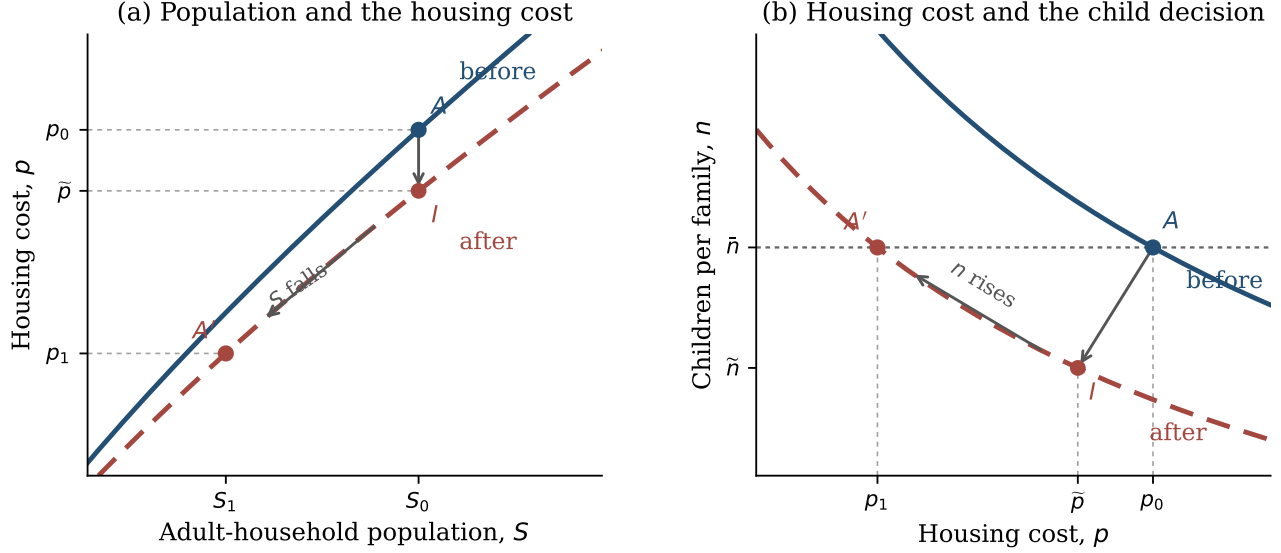


Figure 1: With equal cohort masses, $Y = O = S/2$, housing clearing gives $S^H(p; v) = 2\bar{H}p^\varepsilon / [h_Y(p; v) + h_O(p)]$; these are the left-panel schedules. The child decision determines the steady-state housing cost through $n(p^*) = \bar{n}$. Housing clearing then determines population at that cost. A fall in v initially puts fertility below replacement; as population falls, housing becomes cheaper and partly offsets the initial decline. Blue objects describe the initial value of children and red objects the lower value. Point I and point A' show the impact and new-steady-state comparisons. They do not trace the literal transition, because the ratio Y_t/O_t generally changes away from a steady state.

2.5 Demographic momentum

Let $\mu_t = O_t/Y_t$ be the number of old households per young household, and let $m_t = n(p_t)/\bar{n}$ be the number of next-period young households produced by each current young household. When fertility is below replacement, $m_t < 1$, total household population still rises if

$$\mu_t < m_t < 1. \quad (9)$$

For example, 100 young and 70 old households become 80 young and 100 old households when $m_t = 0.8$. Population rises from 170 to 180 even though the young cohort is not replacing itself.

If preferences and housing supply do not change again between t and $t + 1$, the housing cost also rises when

$$(1 - \mu_t)h_O(p_t) > (1 - m_t)h_Y(p_t). \quad (10)$$

The smaller young cohort reduces demand by $(1 - m_t)Y_t h_Y(p_t)$. At the same time, the current young cohort replaces a smaller old cohort and raises old demand by $(1 - \mu_t)Y_t h_O(p_t)$. The price rises when the second change is larger.

Starting from an exact steady state, $\mu_t = 1$. In this closed two-cohort model, a fertility decline lowers the housing cost on impact and lowers population in the next period. A period in which fertility is below replacement while population and housing costs rise therefore requires an inherited cohort imbalance, produced by earlier fertility, immigration, or both.

On impact, the fertility decline lowers the housing cost relative to the no-shock allocation. Demographic momentum can nevertheless make the realized cost rise over time. The sign comparison above does not by itself order the entire shocked and no-shock price paths.

2.6 What carries to the lifecycle model

The simple model separates two dates. A child changes the parent's need for housing immediately, but changes the number of adult households only after the child forms a household. The lifecycle model retains that distinction, along with tenure constraints, moving costs, saving, and old-age housing retention. Renormalizing the population after every solution would preserve the first effect and remove the second.

3 Quantitative implementation

3.1 The existing household model

The household block is the paper's sequential-fertility model. Households choose saving, tenure, housing size, and whether to attempt a first or a later birth. The state distinguishes children ever born from children currently at home, and retains the owner-size ladder, down-payment constraints, moving costs, income risk, and a warm-glow bequest motive. The population extension does not replace this problem; it changes how the masses generated by the problem enter subsequent dates.

The quantitative specification contains two additions to the earlier household block. The five-state income process is crossed with a three-node discretization of permanent income. Its log variance, 0.393053, is estimated from the PSID and is not a free parameter. We also allow a one-time utility cost F_1 when the first child arrives. Here j is the age index and x collects the remaining household state variables. If $V_j^1(x)$ is the value after a successful first birth, $V_j^0(x)$ is the value after an unsuccessful attempt, and π_j is the conception probability, the value before the attempt-specific taste shock is

$$V_j^A(x) = \pi_j[V_j^1(x) - F_1] + (1 - \pi_j)V_j^0(x). \quad (11)$$

Thus the cost is paid once, conditional on becoming a parent. Setting $F_1 = 0$ recovers the previous fertility problem.

3.2 From births to future households

One model period is four years. Let $\tilde{b}_k$ be births during model period k , measured in child units. Because the parity state is top-coded at 3+, we assign families in that state their observed mean of 3.602 children. We then convert children into future households using

$$B_{k+5} = \frac{\tilde{b}_k}{2.1}. \quad (12)$$

Births during k enter a four-slot queue at the end of the period and first affect the adult distribution at $k + 5$, twenty years later. The number 2.1 is an external replacement normalization: it combines sex composition, survival to household-entry age, and household formation. The model does not contain an explicit childhood-death state. In the constructed old steady state, $\tilde{b}/E = 2.1$, so the mature flow B equals the required entrant flow E .

At a stationary asset price P , let $E(P)$ be required adult entry per adult household, $B(P)$ locally produced mature entry, and $D(P)$ housing demand per adult household. A positive closed steady state requires

$$\frac{B(P)}{E(P)} = 1, \quad SD(P) = H^s(P), \quad (13)$$

where S is the mass of adult households. This is the lifecycle counterpart of replacement in the simple model.

For comparison, suppose that M entrants arrive from outside each period and that a fraction ρ of locally produced entrants remains. An open steady state solves

$$SE(P) = M + \rho SB(P), \quad SD(P) = H^s(P), \quad S = \frac{M}{E(P) - \rho B(P)}. \quad (14)$$

For $M > 0$ and $E(P) - \rho B(P) > 0$, this closure can sustain a positive population below local replacement. In the reported sensitivity, 0.169 is used once to normalize the old steady state: E_0 and B_0 are its required and locally produced entrant flows, $M = 0.169E_0$, and $\rho = (1 - 0.169)E_0/B_0$. Thereafter M and ρ are held fixed, so the realized outside-origin share can change. The normalization is not identified by the 2007–2023 calibration and is not used to improve its fit.

3.3 A calibration along the 2007–2023 transition

For every candidate parameter vector, we first choose an old fertility intercept ψ_{old} so that completed fertility is 2.1. We then preserve the old steady state’s wealth, income, tenure, and family states conditional on age, but replace its age marginal with the national ACS distribution of household heads in 2007. The constructed steady state therefore supplies the histories behind the initial condition; the age-reweighted 2007 economy is already a transition state.

The additive utility weight on children, ψ_t , follows the deliberately simple path

$$\psi_t = \psi_{2007} + \frac{t - 2007}{2023 - 2007} \Delta\psi, \quad t \in \{2007, 2011, 2015, 2019, 2023\}. \quad (15)$$

The change $\Delta\psi$ is estimated, but linearity and the starting date are imposed. The shifter summarizes changes in the value or cost of children not otherwise present in the model; it is not an estimated causal preference shock.

At the same five dates, Census household totals and ACS householder-age shares are imposed exactly. This bridge represents household formation and migration that cannot be generated by births inside a sixteen-year window. It ends after 2023. The historical calibration therefore asks whether the household model, conditional on the observed demographic state, can match the 2023 fertility, housing, tenure, and wealth targets. It does not claim to explain the observed growth or aging of households before 2023.

Housing clears at every date against a constant-elastic supply schedule with elasticity 1.75. After the 2007 age reweighting, its intercept is multiplied by 1.011446 so the inherited price clears the reweighted initial market; the elasticity is unchanged. Households treat the current fertility intercept and current prices as permanent when making decisions. The computation is a sequence of temporary equilibria with an evolving distribution and birth queue, not a perfect-foresight asset-price transition. The model has no separate market-rent path: its asset-price-to-implicit-user-cost ratio is constant under the maintained user-cost identity.

We estimate the ten-element parameter vector ϑ with twelve moments evaluated along the simulated transition. Let $m_k^{2023}(\vartheta)$ denote model moment k , $\hat{m}_k$ its empirical target, and w_k its calibration weight. The objective is

$$Q(\vartheta) = \sum_{k=1}^{12} w_k \left[m_k^{2023}(\vartheta) - \hat{m}_k \right]^2. \quad (16)$$

Completed fertility and childlessness are model age-42 stocks matched to CPS women ages 40–44. The timing moments use a synthetic cohort that combines steady-state prehistory with the 2007–2023 first-birth hazards, and the first-birth rooms moment uses the 2019–2023 branch described below. The remaining moments are evaluated on the simulated 2023 cross-section. Six targets have documented empirical standard errors or stability scales, five use declared provisional calibration scales, and one

is an external normalization. The target documentation records the source, sample, estimator, and uncertainty classification for each row. The criterion is a weighted calibration loss, not a test statistic. The first-birth housing target is provisional. In the PSID it is the change in occupied rooms between event times -1 and $+3$ around a first biological birth. Its model analogue takes the same childless 2019 risk set, branches it into birth and no-birth states, advances both branches one four-year period, and compares their 2023 housing. The Census age bridge is excluded from this comparison. The empirical pre-trend is not flat, so this row remains provisional rather than a clean causal estimate.

Table 1: Calibration along the transition: complete 2023 target fit

Moment	Target	Model	Gap	Weight	Loss
Completed fertility, women 40–44	1.918000	1.898987	−0.019013	1,425.739	0.515
Childless share, women 40–44	0.188000	0.176567	−0.011433	17,180.744	2.246
Mean age at first birth	26.044627	26.338512	0.293885	44.444	3.839
Share of first births at age 30 or later	0.260327	0.242820	−0.017508	10,000.000	3.065
Four-year rooms contrast at first birth	0.720246	0.273250	−0.446997	137.565	27.486
Rooms gap, 3+ versus 1–2 children	0.367700	0.381988	0.014289	2,958.515	0.604
Family ownership gap	0.167662	0.169653	0.001991	14,229.591	0.056
Ownership rate, ages 30–55	0.575472	0.555126	−0.020346	1,207.846	0.500
Mean occupied rooms, ages 18–85	5.779970	6.741957	0.961987	11.973	11.080
Wealth / annual gross labor earnings	6.873100	6.804305	−0.068795	6.288	0.030
Annual bequest flow / aggregate wealth	0.008800	0.008575	−0.000225	5,165,289.256	0.262
Old wealth-to-income p_{90}/p_{50} , ages 76–84	3.448111	3.311776	−0.136335	56.960	1.059
Total loss					50.742

Notes: Contributions use unrounded values and sum to 50.741967. “Family” denotes households with dependent children. The first-birth rooms row, four ACS housing rows, and their declared scales remain provisional; Table values do not conceal that classification.

Table 2: Estimated transition parameters

Object	Estimate	Search bound	Status
Annual discount factor β_{annual}	0.995305	[0.9400, 0.9995]	Estimated; interior
First-birth logit scale κ_f	2.149553	[0.02, 50]	Estimated; interior
Continuation-birth logit scale κ_f^C	1.682038	[0.02, 50]	Estimated; interior
Owner-service premium χ	1.033945	[0.10, 5]	Estimated; interior
Housing-supply scale H_0	16.967273	[0.20, 80]	Estimated; interior
Bequest scale θ_0	0.548015	[0, 8]	Estimated; interior
Bequest shift θ_1	0.100631	[0.02, 16]	Estimated; interior
Child room floor $\bar{h}_c$	0.314216	[0.10, 1.80]	Estimated; interior
First-birth fixed cost F_1	3.793328	[0, 8]	Estimated; interior
Preference change $\Delta\psi$	−0.365051	[−1.50, 0.20]	Estimated; interior

The old normalization gives $\psi_{2007} = 0.230149$ and completed fertility 2.100007; the estimate implies $\psi_{2023} = -0.134901$. No free parameter lies within two percent of a bound in the transformed coordinates used by the search; on the untransformed interval, the bequest shift θ_1 is close to its lower bound. Two independent evaluations of the selected candidate reproduce every moment and the loss to machine precision. The weighted local Jacobian has rank ten at the predeclared relative singular-value threshold 10^{-3} , with condition number 135.6. This is a local numerical diagnostic, not a claim of global identification; the tenure premium and bequest-shift directions are especially non-smooth or weak.

The fit is informative about where the model fails. Fertility stocks, the family ownership gap, and the wealth aggregates are close. The two main misses are the four-year first-birth housing contrast and mean rooms, which together contribute 38.57 of the total loss. Thus the calibrated point should not be described as matching the housing response to fertility.

3.4 What is and is not matched between 2007 and 2023

The historical comparison separates inputs from validation. Household totals and the age distribution are inputs and match by construction. Period fertility, first-birth timing, ownership, rooms, and the

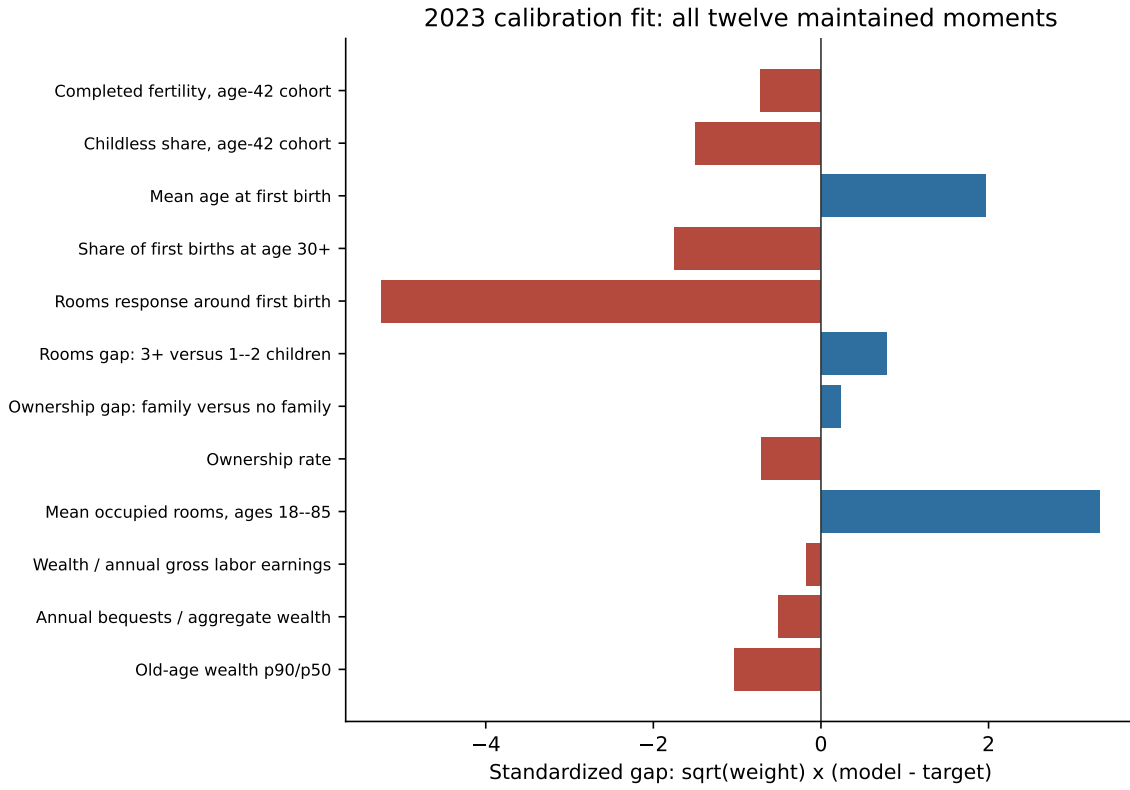


Figure 2: Standardized gaps for all twelve targets. Bars are model minus target, scaled by the square root of the reported weight. The figure reports the same rows as Table 1; it does not convert provisional scales into sampling standard errors.

observed house-price and rent series are not targeted as paths. The 2023 calibration targets are also not uniformly national 2023 cross-sections: several housing targets pool matched metropolitan markets over 2012–2023. The historical national series are therefore holdouts, not hidden calibration rows.

The model captures part of the rise in first-birth age: by 2023 its period mean is within 0.03 years of the NCHS series. It understates the 2023 share of first births at age 30 or later by 2.3 percentage points. Aggregate ownership is within 0.4 percentage points of the national ACS in 2023, but the model misses the fall through 2015 and subsequent recovery. Mean rooms moves in the wrong direction and remains 0.83 room above the 2023 national holdout. The model’s explicit-state and top-bin-adjusted births-per-adult series also decline more sharply than observed period TFR, although those objects are not directly comparable.

The model house-price index rises from one in 2007 to 1.064 in 2023, against 1.156 for real Case–Shiller. More importantly, it cannot reproduce the 2007–2011 collapse and rebound. Its asset-price-to-implicit-user-cost ratio is flat by construction. These misses are a limitation of the temporary-equilibrium price closure, not evidence in favor of a particular fertility shock path.

3.5 Does the calibrated economy have a lower closed steady state?

The theory says that cheaper housing restores a positive closed steady state only if it raises fertility enough to satisfy $B(P)/E(P) = 1$. We evaluate the estimated terminal preference at 25 log-spaced prices from 10^{-4} to three times the old price; denote this grid by $\mathcal{G}$. Required entry $E(P)$ is constant to numerical precision. The renewal ratio is nonincreasing on the audited grid and never crosses one there:

$$\max_{P \in \mathcal{G}} \frac{B(P)}{E(P)} = 0.677985, \quad \min_{P \in \mathcal{G}} \frac{B(P)}{E(P)} = 0.552425. \quad (17)$$

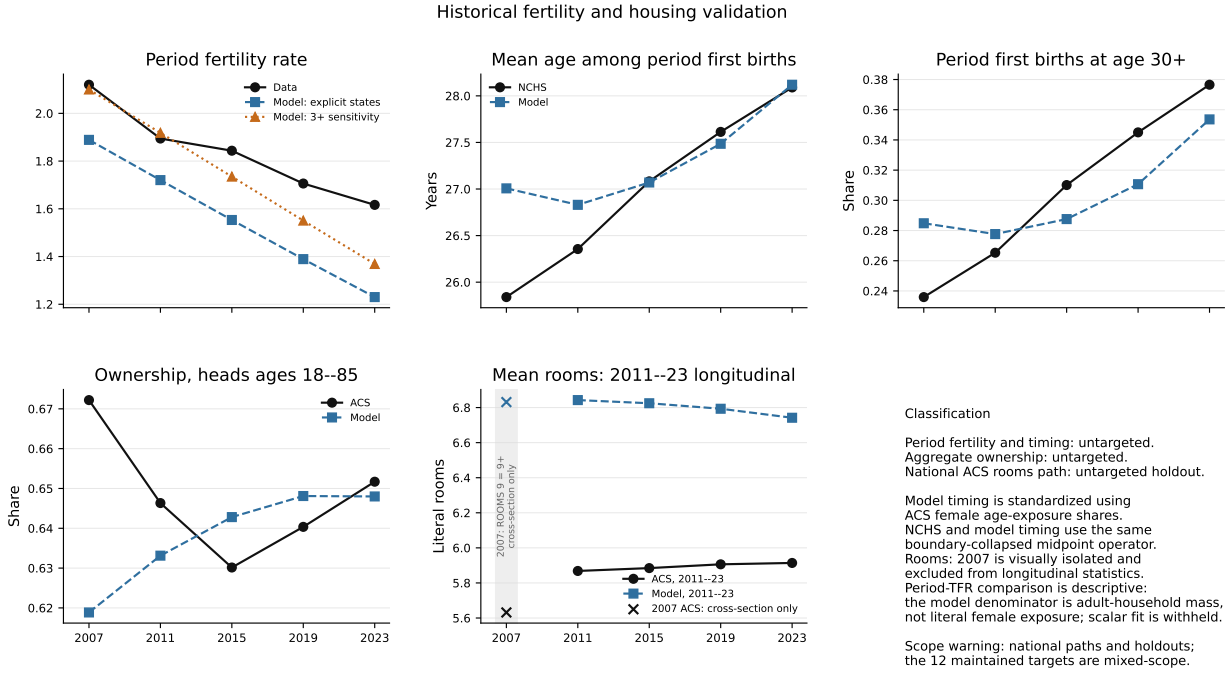


Figure 3: Untargeted demographic and housing-quantity validation. The period fertility series use different denominators and are shown as a descriptive comparison. The 2007 rooms observation is isolated because the ACS top-codes rooms differently in that year.

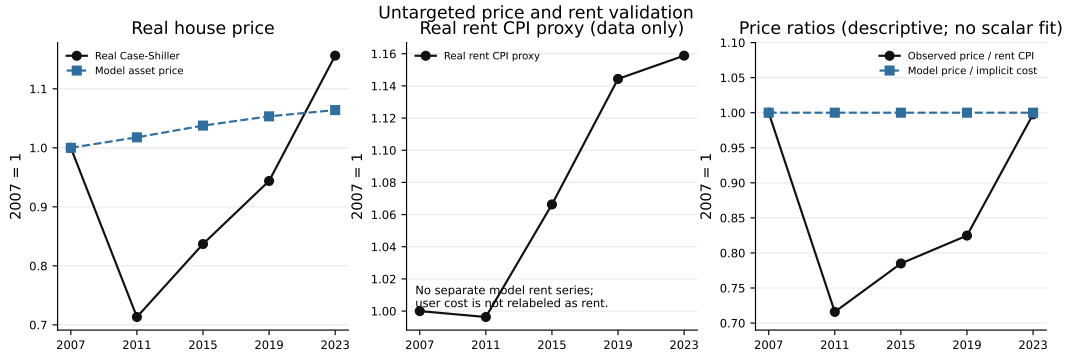


Figure 4: Untargeted housing-price validation. The model reports its temporary- equilibrium asset-price index and the user cost implied by the stationary user-cost identity. Real Case-Shiller and real rent are external comparison series. The model does not have an independent market-rent path.

The maximum occurs at the smallest price and corresponds to top-bin-adjusted completed fertility of 1.424. The schedule is already nearly flat near zero, so the result is not driven by a bad root bracket; the sampled values also provide no evidence of a crossing hidden by a discrete-tenure jump. Using 2, rather than 2.1, as the replacement divisor would raise the maximum renewal ratio only to about 0.712. This is the central quantitative qualification. The simple model's lower positive steady state is a conditional theoretical result; the current calibration does not verify a usable positive root over the audited range. Housing remains a stabilizing feedback, but it does not restore replacement on that grid. The closed post-2023 path can therefore be reported over a finite horizon, not described as movement between two verified positive steady states.

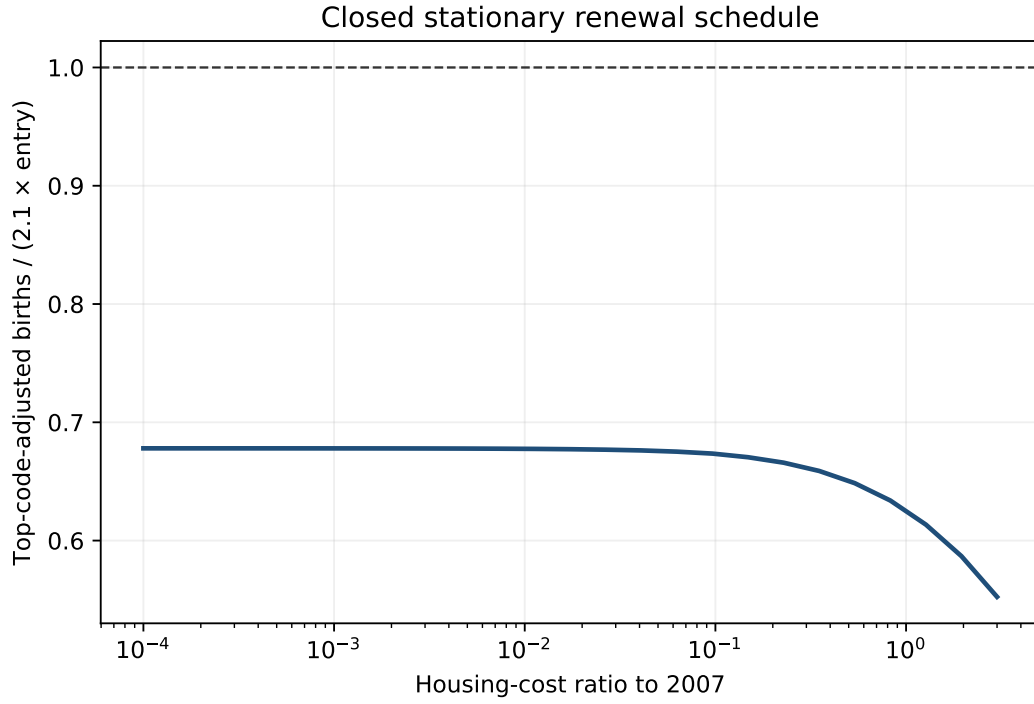


Figure 5: Closed stationary renewal schedule at the estimated 2023 fertility preference. A positive closed steady state requires a crossing of one. No crossing occurs on the audited price grid.

3.6 Paired no-policy continuations after 2023

We advance the exact same 2023 distribution and birth queue under two demographic closures. The national closed benchmark sets $M = 0$ and $\rho = 1$. The open sensitivity holds the externally normalized flow $M = 0.169E_0$ and its associated retention rate fixed. There is no policy or fiscal change. Both paths keep the terminal fertility preference fixed and use the same household and housing-supply blocks.

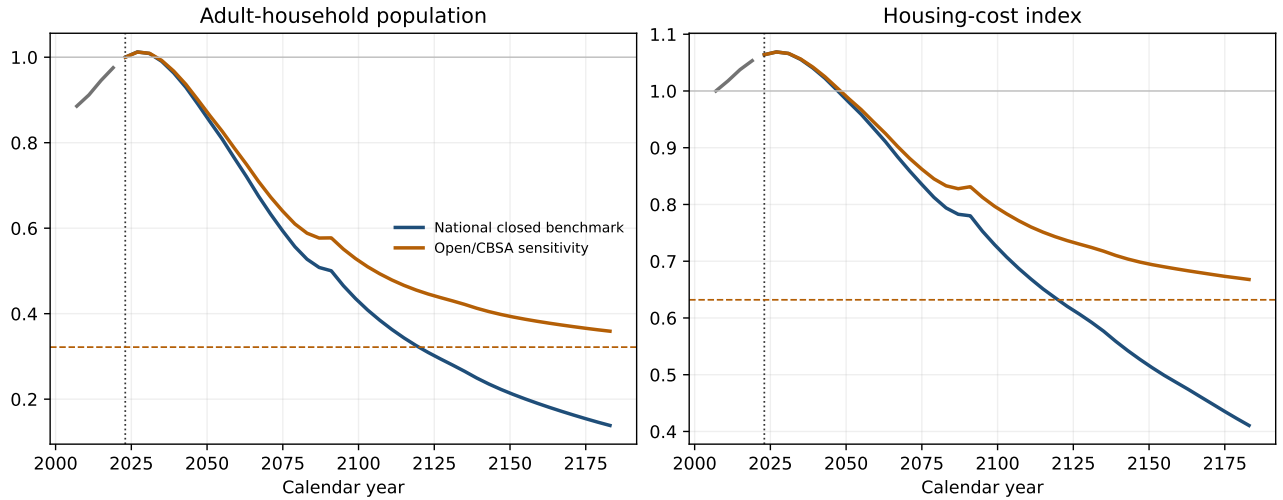


Figure 6: No-policy continuations from the common 2023 state. The closed path is a finite-horizon national benchmark; the open path is a sensitivity to a fixed outside entrant flow. Neither is a forecast.

Both paths initially inherit demographic momentum. In the closed benchmark, adult-household mass rises by 1.24 percent between 2023 and 2027 even though terminal fertility is below replacement; the cohorts entering in 2027 were already born. New post-2023 births first affect entry twenty years later. The paths then separate as outside entry accumulates and the inherited queue is replaced by cohorts born under the lower fertility preference.

By 2183, adult-household mass is (0.138) of its 2023 level in the closed benchmark and (0.359) in the open sensitivity. The corresponding asset prices are (0.386) and (0.628) of their common 2023 value. These are finite-horizon outcomes, not stationary endpoints or forecasts.

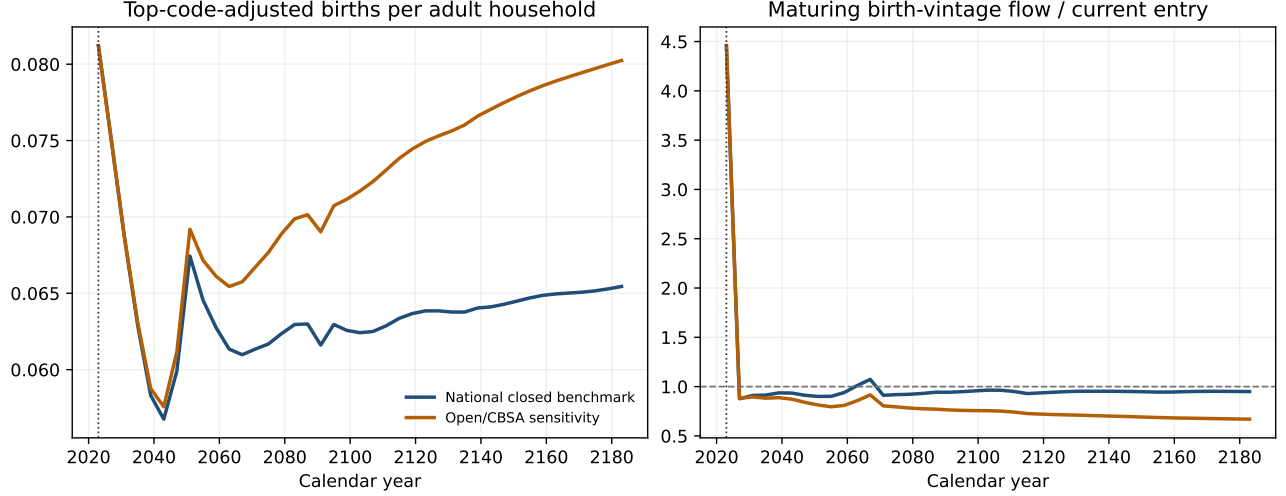


Figure 7: Births, mature entrants, and replacement diagnostics along the paired continuations. Ratios that divide current mature entry by current required entry are transition diagnostics; the stationary root test is reported separately in Figure 5.

The open stationary sensitivity has a population scale of 0.363 and an asset-price ratio of 0.632, both relative to the constructed old steady state. Those levels are conditional on the external old-state normalization and the resulting fixed values of M and ρ ; the realized outside-origin share is endogenous away from that old state and equals 0.465 at this endpoint. They are not identified national forecasts. No corresponding positive closed endpoint is verified over the audited range.

The quantitative result is therefore narrower than a closed-to-closed transition. The exercise establishes a dated 2023 calibration, shows how the existing intergenerational model propagates cohort size, and measures the failure of housing feedback to restore replacement at the estimated point. A causal housing-cost-to-fertility elasticity is the missing moment needed to judge whether re-estimation of the existing fertility block could overturn that conclusion. Adding a steeper response without such evidence would simply engineer the desired root.

4 Conclusion

The paper adds a population law to the existing sequential-fertility and housing model. Children affect their parents' housing immediately, while birth cohorts generate entrant households twenty years later using the replacement conversion in equation (12). This distinction makes the transition intergenerational without assigning estates to particular descendants or replacing the underlying household problem.

The simple theory gives a threshold, not an automatic result. Cheaper housing can return fertility to replacement and produce a lower positive steady state, but only when the fertility response is sufficiently strong. The quantitative model is calibrated at a dated 2023 transition state. Household totals and age shares before 2023 are transparently imposed; the remaining historical series are reported as holdouts. The fit is strongest for fertility stocks, tenure, and wealth, and weakest for the first-birth rooms response and the level of housing consumption.

At the estimated terminal preferences, locally produced entry reaches at most 0.678 of replacement on the audited price grid. No usable positive closed terminal root is verified over that range. Its post-2023 path is a finite-horizon benchmark, while the positive open endpoint is conditional on an externally fixed entrant flow. Neither path is a forecast, and the model's temporary-equilibrium price closure cannot explain the observed house-price crash and rebound or movements in the observed price-to-rent ratio.

A central empirical requirement remains: estimate the causal response of fertility to housing costs or access and use it to discipline the slope of the renewal schedule. If that evidence supports a substantially steeper response, the existing model may deliver a genuine closed-to-closed transition after re-estimation. Without it, adding a steeper fertility response would impose the desired steady state rather than establish it.

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